KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5072 | 5 Credits: 4 | Program Core | Source-grounded

Process Control & Instrumentation

 To impart the basic concepts of process control and instrumentation in process

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5072 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5072.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	5072
Course title	Process Control & Instrumentation
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L: 3 T: 1 P: 0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the basic concepts of process control and instrumentation in process

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding industries. Illustrate the measuring instruments used in	See official syllabus
CO2	19 Understanding process plants. Summarise process control system and final	See official syllabus
CO3	19 Understanding control element.	See official syllabus
CO4	Illustrate analytical instruments. 10 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the elements of process control in industries	4	As specified
Module 2	Illustrate the measuring instruments used in process plants	5	As specified
Module 3	Summarize process control system and final control element	4	As specified
Module 4	Illustrate analytical instruments	3	As specified

MODULE 1

Outline the elements of process control in industries

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the elements of process control in industries
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Understanding chemical plants.

2 Understanding chemical plants. is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding chemical plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding chemical plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding chemical plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Understanding chemical plants. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the control system in a chemical plant.

Outline the control system in a chemical plant. is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the control system in a chemical plant. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the control system in a chemical plant. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the control system in a chemical plant. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Outline the control system in a chemical plant. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring

Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the functions of measuring instruments. 3 understanding comprehend the characteristics of measuring should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.

4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding instruments. summarize the importance of process control in chemical plants - requirements of instrumentation and process control - process variables - advantages of instrumentation and process control. control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - outline a simple level control system – outline a simple temperature control system. functions of measuring instruments, signaling, transmitting, registering, indicating and recording – elements of measuring instruments, primary, secondary, manipulating and functioning element. characteristics of measuring instruments - static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - dynamic characteristics, speed of response, fidelity, lag, dynamic error. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Outline the elements of process control in industries

             Summarize the importance of process control in
  M1.01                                                                 2
Understanding
             chemical plants.

  M1.02      Outline the control system in a chemical plant.            3
Understanding

  M1.03      Outline the functions of measuring instruments.            3
Understanding
             Comprehend the characteristics of measuring
  M1.04                                                                 4
Understanding
             instruments.

Contents:
Summarize the importance of process control in chemical plants - Requirements of
instrumentation and process control - process variables - Advantages of
instrumentation
and process control.
Control system- block diagram - controlled variable, measuring element, measured
value,
controller, set value, error / deviation, desired value, final control element -
Outline a
simple level control system – Outline a simple temperature control system.
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Illustrate the measuring instruments used in process plants

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the measuring instruments used in process plants
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

4 Understanding plants. Summarize pressure measurement in process

4 Understanding plants. Summarize pressure measurement in process is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding plants. Summarize pressure measurement in process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding plants. summarize pressure measurement in process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding plants. Summarize pressure measurement in process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding plants. Summarize pressure measurement in process □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding plants.

4 Understanding plants. is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding plants. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize flow measurement in process plants.

Summarize flow measurement in process plants. is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize flow measurement in process plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize flow measurement in process plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize flow measurement in process plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Summarize flow measurement in process plants. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in

Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize level measurement in process plants. 2 understanding summarize fluid property measurements in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers –Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer.

4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers –Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers –Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding process plants. temperature measuring instruments - principle and working of – thermometers, filled system and vapor pressure thermometers –thermocouples, types, materials - resistance thermometers, thermistor – pyrometer, radiation and optical. pressure measuring instruments - principle and working of - bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. flow measuring devices: coriolis mass flow meter, ultra sonic flow meter, and positive displacement flow meters – differential pressure transducer used in venturi and orifice meters. level measuring devices: float level indicator, magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. ph measurement, ph scale, reference and measuring electrode - conductivity measurement, electrical conductivity meter - density measurement, online density meter, hydrometer. humidity measurement, dew point method, wet bulb method, hygrometer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers –Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure

Aspect	What to prepare
	transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers –Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the measuring instruments used in process plants

             Summarize temperature measurement in process
  M2.01                                                               4
Understanding
             plants.
             Summarize pressure measurement in process
  M2.02                                                               4
Understanding
             plants.
  M2.03      Summarize flow measurement in process plants.             4
Understanding

  M2.04      Summarize level measurement in process plants.            2
Understanding
             Summarize fluid property measurements in
  M2.05                                                                4
Understanding
             process plants.
             Series Test – I                                           1

Contents:
Temperature measuring instruments - Principle and working of – Thermometers,
filled
system and vapor pressure thermometers –Thermocouples, types, materials -
Resistance
thermometers, thermistor – Pyrometer, radiation and optical.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize process control system and final control element

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize process control system and final control element
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Understanding control.

6 Understanding control. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding control. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 6 Understanding control. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize computerized control system.

Summarize computerized control system. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize computerized control system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize computerized control system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize computerized control system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Summarize computerized control system. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize final control elements.

Summarize final control elements. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize final control elements. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize final control elements. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize final control elements. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Summarize final control elements. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks.

Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline instrumented safety system. 3 understanding illustrate on-off (two position control), feedback and feed forward control systems – outline servo and regulatory control - illustrate proportional (p), integral (i), derivative (d) and pid control - illustrate special type of controllers, cascade control, ratio of control and split range control - sketch behaviour of different modes of control. outline analog controllers, electronic controllers and pneumatic controllers. outline computerized control in process plants - programmable logic controllers - supervisory control and data acquisition (scada) - rtu (remote terminal unit) - distributed control system (dcs). final control elements - mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, i to p converter - illustrate valve safe positions - fail open - fail close - fail steady - variable frequency drive [vfd] safety instrumentation system – emergency shutdown system – safety interlocks. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system

Aspect	What to prepare
	(DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize process control system and final control element

             Summarize the principles of automatic process
  M3.01                                                                 6
Understanding
             control.
  M3.02      Summarize computerized control system.                     5
Understanding

  M3.03      Summarize final control elements.                          5
Understanding

  M3.04      Outline instrumented safety system.                        3
Understanding

Contents:
Illustrate On-Off (Two position control), feedback and feed forward control
systems –
Outline servo and regulatory control - Illustrate Proportional (P), integral (I),
derivative (D)
and PID control - Illustrate special type of controllers, cascade control, ratio
of control and
split range control - Sketch behaviour of different modes of control. Outline
analog
controllers, electronic controllers and pneumatic controllers.
Outline computerized Control in process plants - Programmable logic controllers -
Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate analytical instruments

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate analytical instruments
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Summarize chromatographic methods of analysis.

Summarize chromatographic methods of analysis. is an official topic in Module 4 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize chromatographic methods of analysis. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize chromatographic methods of analysis. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize chromatographic methods of analysis. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Summarize chromatographic methods of analysis. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline spectroscopic methods of analysis.

Outline spectroscopic methods of analysis. is an official topic in Module 4 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline spectroscopic methods of analysis. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline spectroscopic methods of analysis. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline spectroscopic methods of analysis. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Outline spectroscopic methods of analysis. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064

Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064 is an official topic in Module 4 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize structural analysis 3 understanding chromatographic analysis – working and applications of - liquid solid chromatography - gas solid chromatography - liquid liquid chromatography - high performance liquid chromatography. spectroscopy – outline the classification of spectrometers – illustrate the working of uv- visible absorption spectrometer - atomic emission spectrometer - mass spectrometer. structure analysis – outline sem (scanning electron microscope), tem (transmission electron microscope), xrd (x-ray diffraction). group activity (prepare a literature review report on the usage of sem, tem and xrd in analyzing nanoparticles) text / reference t/r book title/author t1 outlines of chemical instrumentation & process control; dr. a. sooryanarayana - r1 mechanical& industrial measurements; r.k jain r2 industrial instrumentation ; donald p. eckman - r3 engineering chemistry; jain & jain r4 industrial instrumentation ; k krishnaswami - r5 instrumentation and process control; d.c. sikdar, online resources s.no website link 1 https://www.nptel.ac.in/courses/103105064 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064 □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate analytical instruments

  M4.01      Summarize chromatographic methods of analysis.         3
Understanding

  M4.02      Outline spectroscopic methods of analysis.             3
Understanding

  M4.03      Summarize structural analysis                          3
Understanding

             Series Test – II                                       1

Contents:
Chromatographic analysis – Working and applications of - Liquid solid
chromatography -
Gas solid chromatography - Liquid liquid chromatography - High performance liquid
chromatography.
Spectroscopy – Outline the classification of spectrometers – Illustrate the
working of uv-
visible absorption spectrometer - Atomic emission spectrometer - Mass
spectrometer.
Structure analysis – Outline SEM (scanning electron microscope), TEM
(Transmission
electron microscope), XRD (X-ray diffraction).
Group activity (Prepare a literature review report on the usage of SEM, TEM and
XRD in
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding chemical plants.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding chemical plants.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline the control system in a chemical plant.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the control system in a chemical plant.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead

zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding plants. Summarize pressure measurement in process. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding plants. Summarize pressure measurement in process*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding plants.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding plants.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Summarize flow measurement in process plants.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize flow measurement in process plants.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain 6 Understanding control.. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding control.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize computerized control system.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize computerized control system.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize final control elements.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize final control elements.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Summarize chromatographic methods of analysis.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize chromatographic methods of analysis.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline spectroscopic methods of analysis.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline spectroscopic methods of analysis.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding chemical plants.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding plants. Summarize pressure measurement in process**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **6 Understanding control.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Summarize chromatographic methods of analysis.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- S.No Website Link https://www.nptel.ac.in/courses/103105064

IN-PAGE SEARCH

Search this lesson